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SPECIAL ARTICLE
Heart failure and cardiomyopathies

Prevention of heart failure

A scientific statement of the Heart Failure Association, the European Association of Preventive Cardiology, and the Association of Cardiovascular Nursing & Allied Professions of the ESC, and the ESC Council on Hypertension

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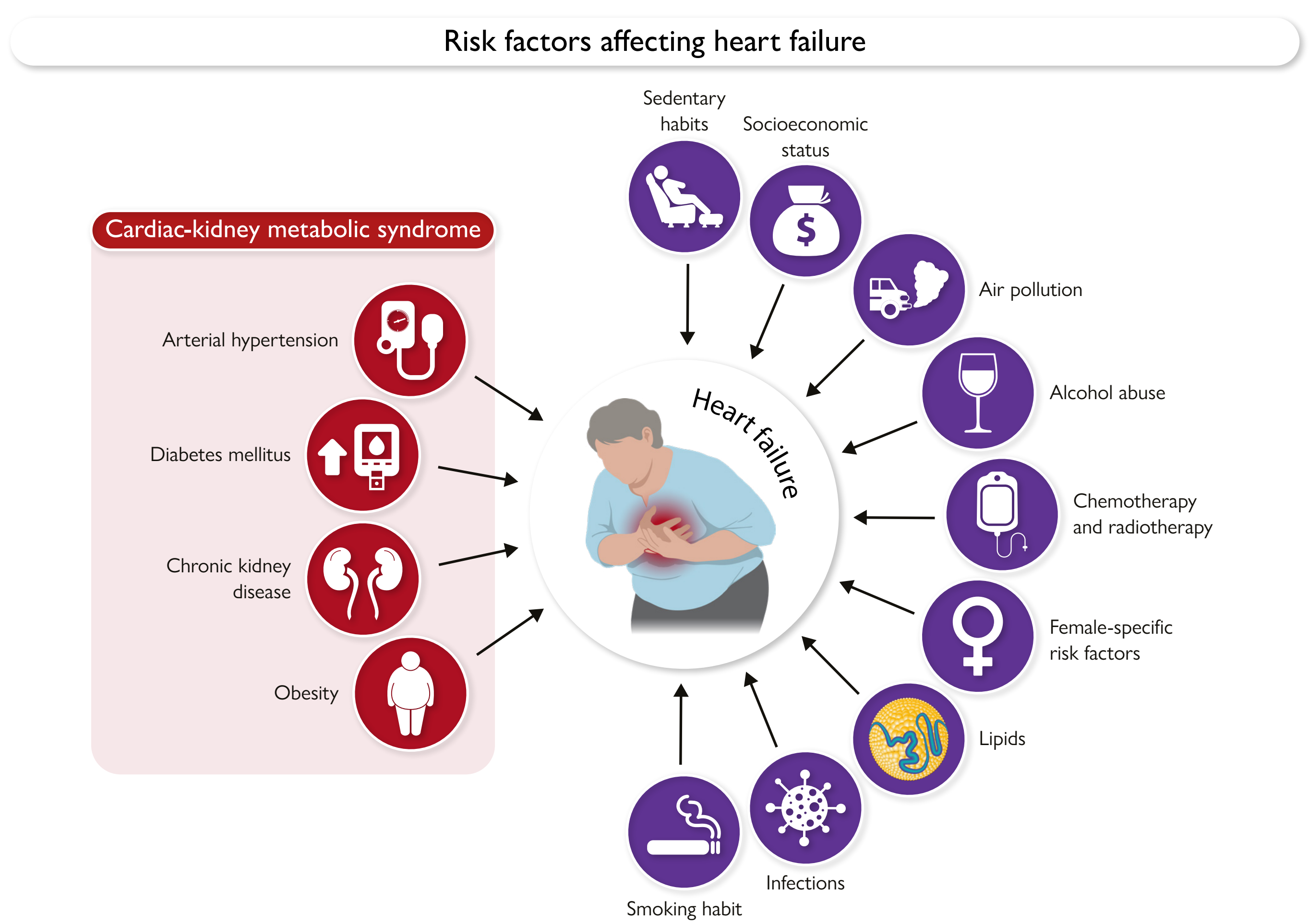
Abstract

Heart failure (HF) remains a major cause of morbidity, mortality and costs for the healthcare systems worldwide, despite advances in diagnostic and therapeutic strategies. The enhancement of preventive measures is now a priority, but effective prevention requires a multidisciplinary strategy addressing a broad spectrum of comorbidities and risk factors. It must also consider the changes in the prevailing phenotype of the patients with HF with a lower impact of coronary artery disease and the increasing role of renal and metabolic conditions leading mostly to HF with preserved ejection fraction. Prevention of HF must take into consideration arterial hypertension, chronic kidney disease, diabetes mellitus, sedentary lifestyle, obesity, dyslipidemia, female-specific risk factors, as well as adverse effects of chemotherapy and radiotherapy. Other key factors include infections and the protective role of vaccination, and environmental and socio-economic determinants of health. In 2022, a position paper of the Heart Failure Association and the European Association of Preventive Cardiology of the ESC was published as a complete overview on this topic and as a compendium to the 2021 ESC Guidelines on HF. However, since then, significant evidence has emerged regarding the potential to prevent HF, particularly in the context of metabolic disorders, diabetes and kidney diseases. This scientific statement aims to provide an updated perspective, highlighting the importance of a holistic and tailored approach to managing the multifaceted contributors to this syndrome.

Graphical Abstract



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Introduction

In 2022, complementing the latest European Society of Cardiology (ESC) Guidelines on Heart Failure (HF), the HF Association, in collaboration with the European Association of Preventive Cardiology, published a position paper offering a comprehensive overview of the prevention of HF.[1] Since then, more robust evidence (*Table 1* and *Table 2*) has emerged regarding the potential to prevent HF with reduced (HFrEF) and preserved (HFpEF) ejection fraction, particularly in the evolving contexts of metabolic disorders and chronic kidney disease (CKD). Consequently, this scientific statement aims to provide an updated perspective, emphasizing the importance of a holistic and tailored approach to managing the multiple risk factors. Preventing HF has a critical role in cardiovascular (CV) medicine, aiming to halt disease progression before clinical symptoms appear.

Screening programmes, including advanced biomarkers and imaging, are effective in early detection and prevention.[52] Lifestyle modifications, structured exercise, and nutritional counselling further reduce HF risk by 25%–30%, along with multidisciplinary management of hypertension, diabetes, and dyslipidemia.[53]

Preventive pharmacotherapy has further advanced with introduction of sodium-glucose co-transporter2- inhibitors (SGLT2i), non-steroidal mineralocorticoid antagonists (MRA) and glucagon-like peptide 1 receptor agonists (GLP1-RA). Comprehensive early intervention and coordinated care strategy offers the most effective means of reducing the global HF burden. *Figure 1* presents some of the more traditional risk factors, while *Figure 2* presents some of the less recognized risk factors for HF and the advised preventive interventions.

This scientific statement is focused on the primary prevention of HF and how to handle its major risk factors. The evidence was identified through a pragmatic and comprehensive review of the available literature. Priority was given to data derived from randomized controlled trials and meta-analyses. Otherwise, findings from prospective and/or retrospective observational studies were also considered. No predefined cut-off date was applied to the literature search.

Epidemiology of HF and risk stratification

HF is a major healthcare concern, marked by high morbidity, mortality, hospitalizations rates, and a significant impact on disability and healthcare cost. European estimates suggest over 14 million prevalent HF cases, with ∼2.5 million incident cases and about 2.2 million HF-related hospitalizations each year.[54] A recent European survey[55] revealed that the median annual incidence of HF per 1000 person-years of 3.9 (IQR 3.1–6.5) and the median prevalence of HF per 100 000 population of 1937 (IQR 1463–3416) patients.

Multiple risk factors contribute to the HF development, and clustering around shared pathophysiological mechanisms, such as systemic inflammation, neuro-hormonal activation, diastolic dysfunction, and metabolic alterations. These mechanisms ultimately lead to HF with both reduced (HFrEF) and preserved ejection fraction (HFpEF).

In clinical practice, it is common to identify clusters of comorbidities (such as obesity, diabetes, and hypertension) which frequently co-exist. CKD and non-traditional risk factors[56] further increase the risk when superimposed on traditional cardiovascular risk factors. An attempt to identify such clusters is represented by the cardiovascular-kidney-metabolic (CKM) syndrome.[57] Early identification of these clusters, through clinical anthropometric, biochemical and imaging evaluations, allows timely treatment and effective primary prevention of HF.

However, the presence of comorbid conditions alone does not fully explain the occurrence of HF. Indeed, it is estimated that only 20%–30%,[58] of patients with arterial hypertension, 12%–22%[59,60,61] of those with diabetes mellitus, and 10%–20%[62] of individuals with obesity will develop HF.

Although an extensive discussion of the pathophysiological mechanisms of HF is beyond the scope of the document, the key factors include endothelial dysfunction, systemic inflammation, neurohormonal activation, coronary microvascular dysfunction, myocardial fibrosis, alterations in cellular metabolism, and oxidative stress.

Risk prediction

A growing body of evidence supports the use of selected biomarkers[63] as valuable tools for predicting the risk of incident HF, independently of the presence of traditional comorbidities.

Neurohormonal activation-such as activation of the renin-angiotensin-aldosterone system and the sympathetic nervous system-occurs in response to disturbances in cardiovascular homeostasis. In general, circulating levels of neurohormones reflect disease severity and have therefore been widely used as biomarkers in patients with HF.

Most useful biomarkers for assessing the risk of developing HF are natriuretic peptides, such as BNP and NT-proBNP[64]; markers of myocardial injury, including high-sensitivity cardiac troponins[65]; markers of impaired renal function such as Urine Albumin-to-Creatinine Ratio (UACR)[66]; markers of systemic inflammation, such as high-sensitivity C-reactive protein.[67] Although several additional biomarkers are under investigation, those reported in *Table 3* currently represent the most widely studied and clinically applied markers for HF risk prediction.

In addition to circulating biomarkers, an elevated resting heart rate has also been identified as a potential predictor of incident HF in asymptomatic individuals, being associated with the development of regional and global left ventricular dysfunction independently of subclinical atherosclerosis and overt coronary heart disease.[69]

SCORE2 and SCORE2-Diabetes are widely used in clinical practice to estimate the 10-year cardiovascular risk of atherosclerotic cardiovascular events (such as myocardial infarction and stroke), these scores have not been validated to estimate the risk of developing HF as a primary event.

Several risk prediction models for HF, such as PCP-HF and PREVENT scores, have been developed and are applicable to the general population without cardiac disease.

The PCP-HF risk calculator accounts sex, ethnicity and is applicable to individuals aged 30 to 79 years without known cardiovascular disease. It was developed using data from multiple cohorts using variables readily available in a primary care setting

Table 1 Common risk factors for heart failure

HF risk	Pathophysiology	Intervention	Evidence
Arterial Hypertension			
Arterial hypertension represents a major risk factor for both forms of HF (HR 1.18 CI 95% 1.13–1.24)[2] HF lifetime risk for 40 years old is 21% in men, and 20.3% in women[3]	Pressure overload that leads to left ventricular hypertrophy and diastolic and/or systolic dysfunction.	Blood pressure target: 120–129/70–79 mmHg, with caution in adults with orthostatic hypotension, moderate-to-severe frailty, limited life expectancy, and older patients (>85 years). The following pharmacologic treatments are advised: thiazide diuretics, ACE inhibitors (ACEi) or angiotensin receptor blockers (ARB), beta-blockers (BB), and calcium channel blockers (CCB). Non-dihydropyridine CCBs may not be as effective for HF prevention compared with other pharmacological agents.	Reduction of SBP to < 120 mmHg was associated with a 38% relative risk reduction in HF rate.[4]
Type 2 Diabetes mellitus			
The relative risk for diabetic patients was 2.06 (CI: 1.73–2.46) with a reported J-shaped association with nadir around 90 mg/dL and increased risk even within the pre-diabetic blood glucose range.[5]	Inflammation, microvascular dysfunction, fibrosis, insulin resistance, lipotoxicity.	Target values for glycaemic control in patients with diabetes mellitus: HbA1c < 7% or < 53 mmol/mol. SGLT2i inhibitors reduce the risk of developing HF in patients with type 2 diabetes at risk of developing cardiovascular disease, as well as those with atherosclerotic cardiovascular disease and those with CKD. SGLT2i are the drug of choice to prevent the onset of HF in these patients. The GLP-1ra have been shown to improve cardiovascular outcomes in patients with type 2 diabetes. In patients with type 2 diabetes mellitus an intensive BP control is beneficial.	• The EMPA-Reg Outcome trial,[6] in patients with atherosclerotic cardiovascular disease, reported a significant 35% reduction in HF hospitalizations (HR 0.65, 95%CI 0.5–0.85) and a similar reduction in CV death or HF hospitalizations (HR 0.66, 95%CI 0.55–0.79). The CANVAS trial[7] with canagliflozin also demonstrated reductions in HF hospitalizations (HR 0.67, 95%CI 0.52–0.87) and CV death or HF hospitalizations (HR 0.78, 95%CI 0.67–0.91). DECLARE-TIMI 58[8] with dapagliflozin and VERTIS-CV[9] with ertugliflozin extended these results. DECLARE reported reductions in HF hospitalizations (HR 0.73, 95%CI 0.61–0.88) and CV death or HF hospitalizations (HR 0.83, 95%CI 0.73–0.95), while VERTIS reported a 30% reduction in HF hospitalizations (HR 0.70, 95%CI 0.54–0.90) but a non-significant 12% reduction in CV death or HF hospitalizations. • In a meta-analysis[10] of 60 080 patients in 8 trials the use of a GLP-1ra led to a reduction in the risk of HF hospitalizations by 11% although this was just statistically significant (95% CI 2% to 18% reduction) • In the BPROAD trial,[11] in which 12 821 patients with type 2 diabetes were enrolled, there was as 21% reduction in the risk of non-fatal stroke, non-fatal myocardial infarction, treatment or HF hospitalizations, or death from CV causes. The point estimate for the

Continued

Table 1 *Continued*

HF risk	Pathophysiology	Intervention	Evidence
			reduction in HF HR = 0.66 (0.41–1.04) was not statistically significant due to the small number of events.
Chronic Kidney Disease CKD is a risk factor for both HFpEF and HFrEF. Both albuminuria and low glomerular filtration rate (GFR), the hallmarks of CKD, are associated with a progressive increase in the risk of major atherosclerotic CV disease, HF events, and CV events, with severe CKD considered as a coronary artery disease risk equivalent.[12,13,14,15,16] Patients with CKD were more likely to develop HFpEF than HFrEF (HR 2.06, 95% CI 1.66–2.56 and HR 1.80, 95% CI 1.36–2.38, respectively).[17]	Endothelial dysfunction, systemic inflammation, neurohormonal activation (RAAS and sympathetic system), accumulation of uraemic toxins, anaemia, alterations in mineral metabolism, vascular and microvascular stiffness, and myocardial fibrosis	SGLT2i prevent HF in patients with diabetic nephropathy to Finerenone prevents HF hospitalization in patients with CKD and type 2 diabetes. Treatment with ACEi/ARB in patients with diabetes mellitus, arterial hypertension, and albuminuria to prevent worsening kidney function. These medications should be titrated to the maximum tolerated dose. GLP1ra (Semaglutide) reduce worsening renal failure in patients with diabetes and CKD. The risk reduction for the composite HF outcome was similar in those with and without HF at baseline, supporting the use of semaglutide in patients with diabetes and CKD to prevent HF.	• Compared with placebo, allocation to an SGLT2i reduced the risk of kidney disease progression by 37%,[18] with similar risk reductions in patients with or without diabetes. The risk of the composite outcome of CV death or HF hospitalizations was reduced by 23%. • In FIDELIO-DKD trial[19] finerenone was associated with a lower occurrence of the key secondary endpoint, including HF hospitalizations (HR 0.86, 95% CI 0.68–1.08). In the FIGARO-DKD trial the benefit was driven by a statistically significant lower incidence of HF hospitalizations with finerenone vs placebo (3.2% vs 4.4%; HR 0.71, 95% CI 0.56–0.90), with no differences in CV death. New-onset HF was significantly reduced with finerenone vs placebo (1.9% vs 2.8%; HR, 0.68, 95% CI 0.50–0.93) • In a *post hoc* analysis[20] of the combined RENAAL and IDNT trials, among patients with diabetes and nephropathy lowering albuminuria and BP in response to ARBs independently correlated with a lower risk of CV outcomes (including HF hospitalizations), suggesting their potential use as therapeutical targets for cardioprotective goals. • In the FLOW trial,[21] 3533 patients with type 2 diabetes mellitus and CKD were enrolled to receive subcutaneous semaglutide at a dose of 1.0 mg weekly or placebo. The risk of a primary-outcome event (major kidney disease events) was 24% lower in the semaglutide group than in the placebo group (HR, 0.76; 95% CI 0.66 to 0.88). In a pre-specified analysis of the FLOW trial, semaglutide reduced the risk of HF events or CV death (HR: 0.73; 95% CI: 0.62–0.87) and of HF events alone (HR: 0.73; 95% CI: 0.58–0.92). The risk reduction for the composite HF outcome was similar in those with and without HF at baseline, supporting the use of semaglutide in patients with diabetes and CKD to prevent HF

Continued

Table 1 Continued

HF risk	Pathophysiology	Intervention	Evidence
Obesity			
Obesity is associated with new-onset HF. This relationship is especially evident for incident HFpEF. Indeed, an entity of 'obesity-related HFpEF' has been described. Tackling obesity prior to the onset of HFpEF (i.e. preventing HF in obese individuals) is an attractive strategy. The relationship between obesity and incident HFrEF is less certain. The cumulative incidence is 1.7% in men and 1.3% in women.[22]	Chronic systemic inflammatory state, endothelial dysfunction, expansion of visceral adipose tissue (including epicardial fat), and altered adipokine secretion, which promote myocardial fibrosis, left ventricular hypertrophy, and ventricular stiffness	Reducing weight in individuals living with obesity can be achieved by lifestyle interventions (education, diet and exercise), Pharmacological approaches Weight loss surgery	The LOOKAHEAD trial[23] showed that the intensive lifestyle intervention did not result in a reduction in HF but resulted in a relatively small between group difference of 4 kg. The investigators reported an association between weight loss and lower HF events. SELECT randomised[24] 17 604 patients living with overweight or obesity (BMI≥27) to semaglutide or placebo. Semaglutide reduced the risk of HF hospitalizations with a similar treatment effect in those with and without HF at baseline (those with HF, HR 0.79 (95% CI 0.64–0.98), those without HF, HR 0.85 (95% 0.68–1.06), *P* interaction 0.64). SURMOUNT-MMO (*n* = 15000, NCT05556512) in people living with obesity is ongoing. There are no randomised controlled trials which have investigated whether or not weight loss surgery reduces HF events. In observational studies, lower HF hospitalizations rates have been reported. A large ongoing randomised controlled trial of weight loss surgery is being conducted in people living with obesity: BRAVE trial (NCT05531474) is recruiting 2000 people living with a BMI ≥35.

ACEi, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BB, beta-blockers; BMI, body mass index; BP, blood pressure; CCB, calcium channel blockers; CKD, chronic kidney disease; CV, cardiovascular; HF, heart failure; HR, hazard ratio; SBP, systolic blood pressure; SGLT2i, sodium-glucose co-transporter 1 inhibitor.

(*Table 4*). A more recent PREVENT equation-based model (*Table 4*) incorporates more variables and allows the inclusion of additional parameters, where available, to improve predictive performance, such as: urine albumin-to-creatinine ratio (UACR) in individuals with CKD, diabetes, or hypertension; HbA1c in individuals with diabetes, prediabetes, increased weight, or a history of gestational diabetes. It includes post code to assess the Social Deprivation Index (SDI), applicable only in the United States. In 2025 an HF risk prediction models was proposed and validated by the European Society of Cardiology Cardiovascular Risk Collaboration: SCORE2-HF[75] (to predict HF irrespective of left ventricular ejection fraction in the general population without cardiovascular diseases).At present, there are no risk scores specifically designed to predict the development of distinct HF subtypes (HFpEF vs HFrEF). The *Table 4* below summarizes the main risk scores currently used for the estimation of HF risk.[70,71,72,73,74,76]

Arterial hypertension

Hypertension affects is a common and the single most important contributor to CV mortality and morbidity. Hypertension is associated with a 3-fold increase in risk of HF in women, and a 2-fold in men.

The association between hypertension and HF has been shown by multiple studies[4] with a 28% increase in risk for each 5 mmHg increase in systolic blood pressure (SBP).[77] The risk of CV disease almost tripled in individuals with BP≥ 130/85 mmHg compared with participants with BP< 120/80 mmHg. A meta-analysis confirmed the crucial role of BP management in reducing HF: on average, a 5 mmHg reduction in systolic BP reduced the risk of major cardiovascular events, with a 13% risk reduction for HF. Furthermore, the presence of hypertension-associated organ damage is associated with an increased risk of HF irrespective of BP level.[78]

Table 2 Other risk factors for heart failure

Risk factors	Interventions
Sedentary habits	• The largest reduction of CV events (including HF events) can be achieved by reaching physical activity levels to 8.75 MET hours per week.[25] Achieving the recommended physical activity in all, would prevent 15.7% (95% CI 13.1–18.2) of all premature deaths. When translating this to the popular 'daily step count', incident CV disease can already be prevented by increasing daily step count from 2000 to 2735 steps per day, with an optimal dose at 7126 steps per day.[26]
Lipids	• Randomized studies, however, failed to show a protective effect of statins against HF[27,28]
Smoking habit	• There was a dose-response relationship for pack-years of smoking and HF. A more extended period of smoking cessation was associated with a lower risk of HF, but risk remained significantly elevated up to 2 decades.[29]
Alcohol, and substance misuse	• Alcohol intake exceeding 70 g/week increases the risk of HF progression by 4.9 times,[30] while cannabis, opioid, and stimulant users exhibit a 2.52-, 3.48-, and 3.35-fold higher HF prevalence, respectively.[31,32]
Chemotherapy and radiotherapy (RT)[33,34,35,36,37,38,39,40]	• Optimize risk factors control (smoking cessation, restriction of alcohol consumption, sedentary habit, etc), according to the ESC Guidelines on CV prevention. • High and very high-risk patients have a guidelines indication to minimize the use of cardiotoxic drugs and should start medication with ACE-I/ARB and Beta-blockers, and statins. • Dexrazoxane and liposomal anthracyclines are currently approved in patients with high and very high risk or who have already received high cumulative anthracyclines dose. • Primary prevention of RT-induced damage depends on technological advances that allow improved targeting of RT delivery, thereby maintaining or increasing oncological efficacy. There is no proven medical therapy to prevent RT-induced CV toxicity. • Future studies are evaluating SGLT2i therapy in preventing CTRCD
Infections	• In a *post hoc* analysis of the PARADIGM-HF trial, of 8099 study participants with HF, 1769 (21%) received influenza vaccination and this was associated with a reduced risk for all-cause mortality in propensity-adjusted models.[41] The INVESTED trial compared the effect of high-dose trivalent and standard-dose quadrivalent inactivated influenza vaccine in patients with recent HF hospitalization or acute myocardial infarction, with no difference observed in risk of death or cardiopulmonary hospitalization between treatment groups.[42]
Female-specific risk factors	• Observational epidemiological studies have also documented the independent association of female-specific reproductive factors with increased risk of HF, including early menarche, early menopause, (i.e. before the age of 45 years),[43] recurrent pregnancy loss, nulliparity,[44] hypertensive pregnancy complications[45] and endometriosis.[46]
Air pollution	• Chronic exposure to $PM_{2.5}$, NO_2, O_3, and O_x is associated with a higher risk of myocardial infarction and HF,[47] even at pollutant levels considered acceptable by current standards.[48,49]
Socio-economic status	• Socioeconomic status strongly influences HF risk, with factors such as food insecurity, poverty, and social isolation playing significant roles[50]
Sleep disorder breathing (SDB)	• Obstructive sleep apnoea predicted incident HF in men but not in women (HR 1.13 per 10-unit increase in AHI): AHI >30 was associated with higher risk to develop HF than AHI <5.[51]

AHI, apnoea/hypopnea index; CTRCD, Cancer therapy-related cardiac dysfunction; NO, nitric oxide; O_3, ozone; PM, particulate matter; RT, radiotherapy. (For the remaining, refer to *Table 1*).

Women seem more affected by arterial hypertension as a risk factor for HF. A total of 91% of the participants in the Framingham Study who developed HF had a previous diagnosis of hypertension, and the risk of HF occurring in the absence of myocardial infarction was 1 in 9 for men and 1 in 6 for women. For all HF events, the population-attributable risk of hypertension, defined as SBP superior or equal to 130 mmHg and DBP superior or equal to 80 mmHg, was 25.5% in black women, 17.3% in white women, 28.3% in black men, 17.3% in white men.[3]

Evidence[79,80,81] from large observational studies and meta-analyses suggests that women may receive less optimal hypertension management than men, with lower use of key cardioprotective agents such as ACE inhibitors and poorer blood pressure control at older ages despite similar treatment rates overall. These sex disparities in prescribing patterns and age-related hypertension control may contribute to the higher risk of hypertension-related heart failure observed in women.

Adequate BP control prevents the development of hypertension-mediated organ damage and progression to overt HF. In the SPRINT trial a target systolic BP < 120 mmHg was associated with a 38% relative risk reduction in HF.[4] In the BROAD randomized clinical trial, the protective role of an

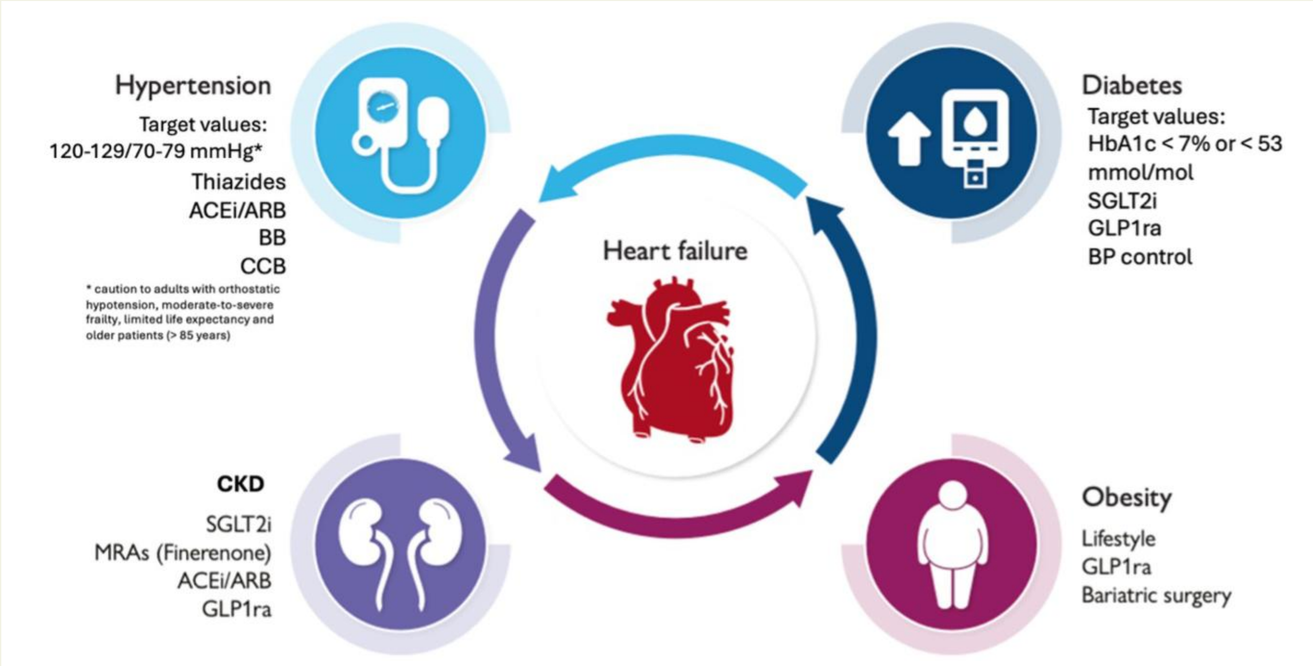

Figure 1 Traditional risk factors for heart failure and advised preventive strategies. ACEi, angiotensin-converting enzyme inhibitors; ARB, angiotensin receptor blockers; BB, beta-blockers (BB); BP, blood pressure; CCB, calcium channel blockers; GLP1ra, glucagon-like peptide 1 receptor agonists; MRA, mineral corticoid antagonists; SGLT2i, sodium-glucose co-transporter2 inhibitors

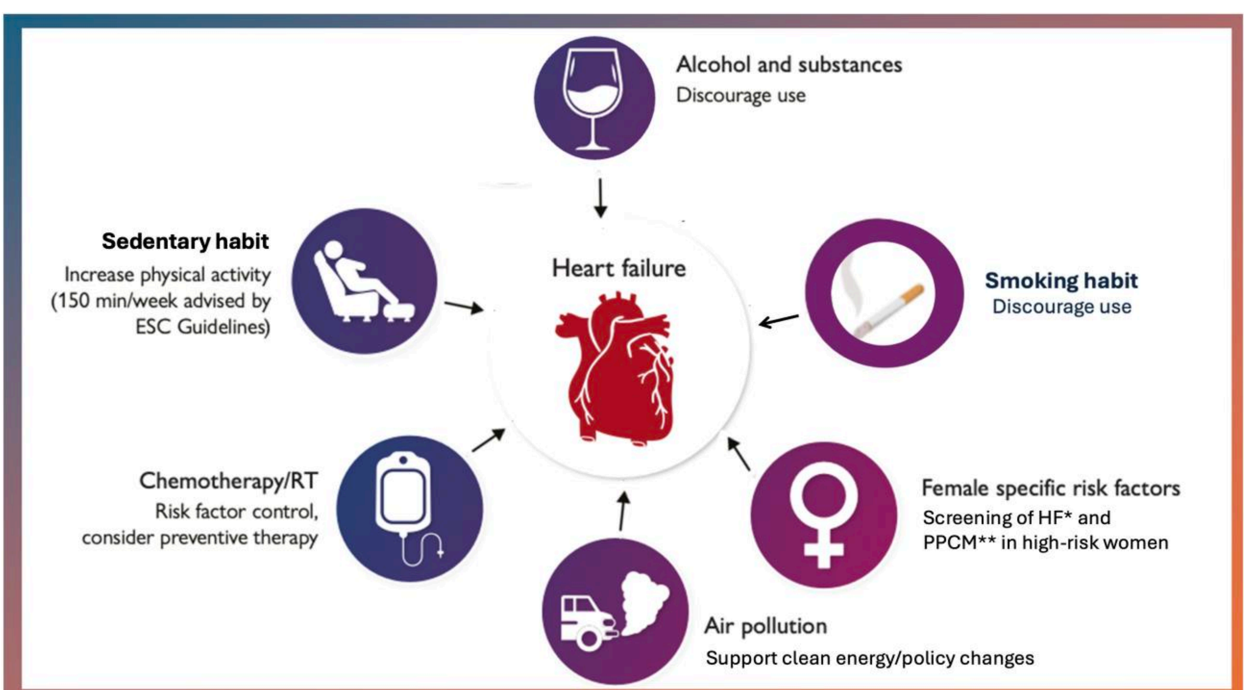

Figure 2 Less recognized risk factors for heart failure and advised preventive strategies. HF, heart failure; RT, radiotherapy. *Early menarche, early menopause, (i.e. before the age of 45 years), recurrent pregnancy loss, nulliparity, hypertensive pregnancy complications and endometriosis. ** Ethnicity (higher incidence in Africans), genetic variance in dilated cardiomyopathy genes, pre-eclampsia, multi-gestational pregnancies, maternal age <20 or >40 years, smoking, diabetes, hypertension and family history of cardiomyopathy

Table 3 The most widely studied and clinically applied markers for heart failure risk prediction

Biomarkers	Pathophysiology	Clinical context	Cut-off
NT-proBNP/BNP	Ventricular wall stress, neurohormonal activation	Values influenced by age, sex, BMI, and renal function. NT-proBNP more influenced than BNP to age and renal function, thus specified cut off are needed.	The following cut off are advised to rule out HF: BNP ≥35 pg/mL, NT-proBNP ≥125 pg/mL[64,68]
Cardiac troponin (c-Tn)	Subclinical myocardial damage	Modest increase in discrimination compared with NT-proBNP. Useful when BNP/NT-proBNP values are ambiguous	hs-cTnI ≥4.2 ng/L (men), ≥2.6 ng/L (women)[65]
Urine Albumin-to-Creatinine Ratio (UACR)	potential kidney damage	It is advised to measure it on a spot urine sample (random or preferably first-morning urine)	A linear relationship between level of albuminuria and relative hazard of the development of HF has been described.[66] Albuminuria is associated with a progressively increased HF risk from intermediate-normal (adjusted HR, 1.54; 95% CI, 1.12–2.11) and high-normal UACR (adjusted HR, 1.91; 95% CI, 1.38–2.66) to microalbuminuria (adjusted HR, 2.49; 95% CI, 1.77–3.50) and macroalbuminuria (adjusted HR, 3.47; 95% CI, 2.10–5.72)[66]
High sensitive C reactive protein (hsCRP)	Systemic inflammation	Dose-dependent risk. Useful in a multimarker panel	hsCRP ≥2 mg/L is associated with MACE in the general population[67]

MACE, major adverse cardiac event. (For the remaining, refer to *Table 1*).

intensive BP–lowering strategy was confirmed in diabetic patients at high cardiovascular risk for a composite endpoint that also included HF [HR 0.79 (0.69–0.90), $P < .001$].[82] However, this study was not designed for HF and therefore was not able to specifically evaluate the effect on the single endpoint of HF. Evidence regarding the benefits of intensive blood pressure treatment for the prevention of HF has also been confirmed by individual patient data meta-analyses.

Many classes of drugs, including angiotensin-converting enzyme inhibitors (ACEi)/angiotensin receptor blockers (ARBs), beta-blockers (BB), calcium channel blockers (CCBs), and diuretics, are available to reduce BP[78] (*Table 1*). Current findings have shown that non-dihydropyridine CCBs seem to be inferior at preventing HF compared with ACEi/ARBs, BB, and diuretics. The protective effect of the ACEi/ARB class against HF may be independent of BP reduction alone, mediated through modulation of the renin–angiotensin–aldosterone system and the sympathetic nervous system. In the HOPE study, which enrolled high-risk patients older than 55 years, treatment with ramipril reduced the rates of death, myocardial infarction, stroke, coronary revascularization, cardiac arrest, HF, diabetes and diabetes-related complications. However, the benefits could only be partly attributed to BP reduction, since most patients did not have hypertension at baseline and the mean reduction in BP with treatment was small (3/2 mmHg). In a comparative study between valsartan and amlodipine (the VALUE study), although amlodipine showed a greater effect in reducing BP, particularly in the early phases of treatment, a trend toward a reduction in HF was observed in the valsartan group, although this did not reach statistical significance.[83] In the ALLHAT study,[84] patients treated with amlodipine showed a higher rate of HF compared with those treated with chlorthalidone and lisinopril, highlighting that among the three drugs, chlorthalidone demonstrated the greatest protective effect against HF. Evidence derived from a meta-analysis confirmed that both thiazide diuretics and ACEi/ARBs exert a statistically significant protective effect against HF compared with non-dihydropyridine CCBs, whereas this effect was only numerically lower when compared with dihydropyridine CCBs.

An ‘ad hoc’ meta-analysis[85] focused on antihypertensive treatment and the HF prevention demonstrated that BP-lowering therapy effectively prevents new-onset HF. Moreover, the findings indicate that BP reduction achieved with CCB is as effective as that obtained with other antihypertensive agents in preventing new-onset HF, except in trials in which the study design introduces an imbalance against calcium antagonists.

Current guidelines[68] do not provide clear recommendations favouring the use of ACEi/ARBs and thiazide diuretics over CCBs as for HF prevention. Further randomized clinical trials will be necessary to determine whether a class effect exists in the prevention of cardiovascular risk for HF.

Diabetes Mellitus

HF is one of the most frequent CV outcomes in patients with type 2 diabetes.[86] Large-scale cardiovascular outcomes trials and real-world comparative effectiveness studies have established that the choice of antidiabetic agent, rather than glycemic control alone, is a key determinant of preventing HF. Recently

Table 4 Risk scores developed for the estimation of heart failure risk

Score	Included Variables	Derivation Population	Application Setting
Health ABC Heart Failure Score	Age; sex; blood pressure; heart disease; smoking; glucose; creatinine; albumin; left ventricular hypertrophy	Health ABC study (older adults aged 70–79 years, without HF)	The Health ABC HF model adequately predicted 5-year heart failure risk[70]
ARIC Risk Score	History of ischaemic heart disease; blood pressure; DM; BMI; cigarette smoking; COPD; ethnicity; ECG characteristics; biomarkers such as NT-proBNP	ARIC	Prediction of heart failure risk in the general population[71]
PCP-HF	Age; sex; ethnicity; blood pressure; glucose; cholesterol; BMI; smoking; QRS duration	5 U.S. cohorts	The PCP-HF risk score predicts CVD and all-cause mortality, in addition to the 10-year risk of incident HF among white and black men and women[72]
PREVENT	Sex; smoking status; use of lipid-lowering therapy; age; HDL cholesterol; BMI; total cholesterol; SBP; eGFR; DM; use of antihypertensive therapy	46 U.S. cohorts; extensively validated	A sex- and race-specific 10-year pooled cohort equations to Prevent HF risk score, which integrates clinical parameters readily available in primary care settings. This tool can be useful in risk-based decision making to determine who may merit intensive screening and/or targeted prevention strategies[73]
Framingham Heart Failure Risk Score	Age; sex; blood pressure; diabetes; heart disease; BMI; ECG variables; smoking	Framingham cohort	Derived from a predominantly White population including both healthy individuals and subjects with cardiovascular disease; application in the general population may overestimate risk in truly healthy individuals[74]
SCORE2-HF	age, smoking status, SBP, antihypertensive treatment, BMI, eGFR, DM	UK Biobank and multiple European and North American cohort	Predict HF irrespective of left ventricular ejection fraction in the general population without cardiovascular diseases. Models were recalibrated to reflect HF incidence in different risk regions of Europe[75]

Abbreviations. Refer to *Table 1*.

the emergence of therapies that can reduce the risk of HF in this population has changed guidelines to make prevention of HF an achievable goal. Until the clinical trials of SGLT2i, no glucose-lowering therapy has been shown to reduce the risk of HF. Studies of metformin were based on observational data or small subgroups[87,88] and the risk of HF was potentially increased by dipeptidyl peptidase-4 (DPP-4) inhibitors.[8,90,91]

Hypertension is a common comorbidity in type 2 diabetes. It is consistently associated with poorer outcomes and trials have assessed intensive compared with lenient BP control in this population. In the BROAD trial[11] an intensive BP lowering strategy, with a target of <120 mmHg, was superior to a standard target one (<140 mmHg) regarding CV outcomes, while, in the ACCORD trial a more aggressive BP control reduced the annual rate of stroke (*Table 1*).

Sodium-glucose co-transporter2-inhibitors (SGLT2i)

The SGLT-2i was the first drugs for the treatment of type 2 diabetes that demonstrated that HF could be prevented in this population.

Many clinical trials (i.e. EMPA-Reg Outcome trial,[6] CANVAS trial,[7] DECLARE-TIMI 58[8] and VERTIS-CV[9]) have reported significant reductions in HF hospitalization and CV mortality with SGLT2i (*Table 1*). SGLT-2i reduce HF hospitalizations by 30% and CV death or HF hospitalizations by 21%, supporting their role in preventing HF in type 2 diabetes[91] (*Table 1*).

Glucagon-like peptide 1 receptor agonists (GLP-1ra)

GLP-1ra have been shown to improve CV outcomes in patients with type 2 diabetes. They reduce the risk of myocardial infarction and stroke in patients with type 2 diabetes, but, most trials using GLP-1ra have not been powered to examine HF outcomes, while reporting HF as a secondary outcome.[10,21,24,92–94]

GLP-1ra lower the risk of HF hospitalizations (*Table 1*) In a meta-analysis of 60 080 patients in 8 trials, the use of a GLP-1ra reduced in the risk of HF hospitalizations by 11% although this was just statistically significant (95% CI 2% 18%).[10] Therefore, the benefits of GLP-1ra to prevent the onset of HF alone would appear to be less certain than the use of SGLT2i although they may be used for primary prevention of

Table 5 Gaps in Knowledge

• The use of GLP-1ra with a SGLT2i to prevent the onset of HF has not been studied. Whether they have an additive benefit in reducing the risk of HF remains unclear.
• In a pre-specified analysis of the FLOW trial, GLP1ra reduced the risk of HF events or CV death and of HF events alone similarly in those patients with and without HF at baseline: further evidence is needed to support the use of this class of drugs to prevent HF in patients with diabetes and CKD.
• Further dedicated pharmacological trials with a primary objective of reducing incident HF in patients living with obesity are warranted.
• The development of future trials of Lp(a) reducing agents may help clarify the true association with HF and determine if HF can be prevented by reducing Lp(a) levels.
• Additional recommended drugs to prevent HF may include statins to reduce the risk of major atherosclerotic events (namely, coronary artery disease), often mediating new-onset HF.
• An important decline in physical activity levels worldwide, associated with an increased risk of HF has been described, but effective interventions to reverse this ominous trend are lacking.
• The role of bariatric surgery in preventing HF is still unknown.
• The mechanisms of HF associated with substance misuse are diverse and not fully understood (including genetic predisposition).
• In CTR-CVT, the role of pharmacological therapy in preventing HF warrants further investigation, including the role of SGLT2i.
• Sleep disordered breathing (SDB) is associated with increased risk of CV diseases and HF, but there is still no evidence that treating SDB prevent HF occurrence.
• In women, the mechanisms underlying the association between reproductive factors and HF are unclear: these associations were attenuated when adjusted for the presence of metabolic risk factors like body mass index, type 2 diabetes and systolic blood pressure. The role of oestrogen replacement therapy in post-menopausal women remains controversial. Whether the initiation of BP lowering treatment at lower BP level in women with reproductive risk factors may improve the prevention of HF.
• Against air pollution, studies are needed to assess the efficacy of interventions in reducing HF risk.
• The efficacy of interventions targeting HF patient care stopping progression of the syndrome

HF in the patients with type 2 diabetes who require further glucose-lowering.

Other antidiabetic medications

Apart from the above-mentioned medications (i.e. SGLT2 inhibitors and GLP-1 receptor agonists), some glucose-lowering drugs have a neutral effect, whereas others may be harmful in patients with HF or in those at increased risk of HF.[93]

Thiazolidinediones[94] (i.e. pioglitazone, rosiglitazone) may have a negative impact in patients with HF or at high risk, due to fluid retention, both as monotherapy and in combination with other antidiabetic agents.

DPP-4 inhibitors show different impacts based on the molecule considered. Thus, saxagliptin[89,95] increases the risk of hospitalization for HF and it is advised to avoid it in patients at risk, whereas sitagliptin and linagliptin showed neutral effect.[96–98]

Metformin[99] is neutral or potentially beneficial: it does not increase the risk of HF and may be associated with a modest reduction in events. As for sulfonylureas, this class of medication may have a higher risk of HF, especially when compared with metformin.[100,101] However, it should be noted that specific RCT evaluating the impact of sulfonylureas on the risk of developing HF are lacking. RCTs[102,103] with insulin glargine and/or degludec did not demonstrate any effect on HF risk.

In conclusion, an approach focused on the choice of glucose-lowering agents, rather than on glycemic control alone, is crucial for the prevention of HF in patients with diabetes.

Chronic kidney disease

CKD is a strong risk factor for CV disease and HF.[12,13,14] Both albuminuria and low glomerular filtration rate (GFR), the hallmarks of CKD, are associated with a progressive increase in the risk of major atherosclerotic CV disease, HF events, and CV events, with severe CKD considered as a coronary artery disease risk equivalent.[13,14,15,16,104]

Kidney-heart interactions contribute to a reciprocal risk amplification of renal and cardiac diseases due to shared risk factors (e.g. hypertension, diabetes), common pathogenic mechanisms (e.g. renin-angiotensin system activation, inflammation), and advanced CKD complications like vascular calcification, coronary artery disease, and anaemia, which exacerbate HF.[105,106] CKD often coexists with cardiac and metabolic conditions, leading to the definition of the cardiac-kidney-metabolic (CKM) syndrome by the American Heart Association.[107] However, no clear universal definition of CKM has been adopted to date, and the following definition proposed by the American Heart Association is commonly referenced: CKM syndrome is defined as a systemic disorder characterized by pathophysiological interactions among metabolic risk factors, CKD, and the cardiovascular system, leading to multiorgan dysfunction and a high rate of adverse cardiovascular outcomes.[75] In a sub-study of DELIVER trial,[104] which included 6263 patients with HF, 31% had 1 CKM condition, 36% had 2, and 20% had all 3 of them (CKD, type 2 diabetes, and atherosclerotic CV disease). The presence of 3 CKM conditions doubled the risk of primary events compared with HF alone. Routine screening for CKD (via GFR and albuminuria) and other CKM risk factors is needed to prevent and for early management of HF.[108,109] However, evidence is lacking for patients with severe CKD (eGFR <30 mL/min/1.73 m^2), as they are often excluded from clinical trials.

ACE-i/ARB

Treatment with an ACEi or an ARB is advised in patients with diabetes, hypertension, and albuminuria to prevent worsening kidney function and to prevent HF[20] (Table 1). These medications should be titrated to the maximum tolerated dose.

SGLT2i

CREDENCE, SCORED, DAPA-CKD and EMPA-KIDNEY are the major trials testing SGLT2i (canagliflozin, sotagliflozin, dapagliflozin and empagliflozin, respectively) in patients with CKD, with or without diabetes[110–113] (Table 1). Albuminuria was used as an inclusion criteria in these trials. All these trials met their primary endpoint, including reducing HF hospitalizations. After planned interim analyses, CREDENCE and EMPA-KIDNEY trials were stopped early due to the significant reduction of the risk of the primary endpoint with SGLT2i.

A subsequent pre-specified meta-analysis, including 13 trials, enrolled patients with either type 2 diabetes and high atherosclerotic CV risk, with 90 409 participants, of whom 74 804 (82.7%) with and 15 605 (17.3%) without diabetes: compared with placebo, allocation to an SGLT2i reduced the risk of kidney disease progression by 37%, with similar risk reductions in patients with or without diabetes.[18] The risk of the composite outcome of CV death or HF hospitalizations was reduced by 23%. However, results were significant in patients with CKD and diabetes but not in those without diabetes, at least in part due to the small number of events in this group.[18,110,114–116] These findings led to an indication for SGLT2i to prevent HF in patients with diabetic nephropathy in the 2023 Update of the ESC HF guidelines.[68]

Treatment with SGLT2i also mitigated the longitudinal rise in biomarkers' concentrations, including NT-proBNP (*n*-terminal pro-B-type natriuretic peptide), high-sensitivity cardiac troponin T, growth differentiation factor-15, and IGFBP7 (insulin-like growth factor binding protein 7).[110] No data are available for patients with severe CKD, namely patients with eGFR < 20 mL/min/m^2, < 25 mL/min/m^2 or < 30 mL/min/m^2 who were excluded from EMPA-KIDNEY, DAPA-CKD, SCORED or CREDENCE, respectively.[110,111–113]

Glucagon-like peptide 1 receptor agonists

Among glucose-lowering therapies, GLP-1RAs also have established CV disease and renal benefits[10] (Table 1). The recently published FLOW trial randomly assigned 3533 patients with diabetes and CKD to receive subcutaneous semaglutide at a dose of 1.0 mg weekly or placebo.[21] The primary outcome was major kidney disease events, a composite of the onset of kidney failure (dialysis, transplantation, or an eGFR of <15 mL per minute per 1.73 m^2), at least a 50% reduction in the eGFR from baseline, or death from kidney-related or CV causes. Median follow-up was 3.4 years and early trial discontinuation was advised at a pre-specified interim analysis. The risk of a primary-outcome event was 24% lower in the semaglutide group than in the placebo group.[21] Results were significant also for the composite of kidney-specific components of the primary endpoint and for death from CV causes when analyzed separately, with a 21% and 29% risk reduction, respectively.

In a pre-specified analysis of the FLOW trial, semaglutide reduced the risk of HF events or CV death (HR: 0.73; 95% CI: 0.62–0.87) and of HF events alone (HR: 0.73; 95% CI: 0.58–0.92). The risk reduction for the composite HF outcome was similar in those with and without HF at baseline. Though these results are limited by the small number of events, compared with the rates observed in patients with HF, they support the use of semaglutide in patients with diabetes and CKD to prevent HF.[117]

The SELECT trial demonstrated that one-weekly dose of 2.4 mg of subcutaneous semaglutide was associated with a 20% reduction in major adverse CV events vs placebo in patients with pre-existing CV disease and a BMI of ≥27 kg/m^2, without diabetes.[118] In a pre-specified analysis of this trial, semaglutide was associated with a 22% reduction in the main 5-component kidney composite endpoint, with benefits mainly driven by reduced incidence of macroalbuminuria and onset of persistent ≥50% reduction in eGFR. The treatment benefit at 104 weeks for eGFR slope was 0.75 mL/min/1.73 m^2 (*P* < .001) overall and 2.19 mL/min/1.73 m^2 (*P* < .001) in patients with baseline eGFR <60 mL/min/1.73 m^2.[24]

In a small trial, including 101 patients, semaglutide treatment 2.4 mg per week for 24 weeks resulted in a clinically meaningful reduction in albuminuria in patients with overweight/obesity and non-diabetic CKD as compared to placebo, but further larger trials are needed in this regard.[92]

Mineralcorticoidreceptor antagonist

The selective, non-steroidal, MRA finerenone was tested in two trials in patients with diabetic kidney disease, the FIDELIO-DKD (*n* = 5734) and FIGARO-DKD (*n* = 7437) trials (Table 1).

In the FIDELIO-DKD trial, eligible patients had a uACR ratio 30–300 mg/g, an eGFR 25–60 mL/min/1.73 m^2, and diabetic retinopathy, or a uACR 300–5000 mg/g and an eGFR 25–75 mL/min/1.73 m^2. The primary outcome, a composite of kidney failure, a sustained decrease of ≥40% in the eGFR from baseline for more than 4 weeks, or death from renal causes, assessed in a time-to-event analysis, was significantly reduced by finerenone.[19] Cardiorenal outcomes were improved irrespective of baseline HF history.[111] In FIDELIO-DKD finerenone was associated with a lower occurrence of the key secondary endpoint including HF hospitalizations (HR 0.86, 95% CI 0.68–1.08).[19]

FIGARO-DKD trial enrolled patients with stage 1 or 2 of CKD and eligible patients were treated with renin-angiotensin system blockade at maximum tolerated dose.[112] The primary outcome, assessed in a time-to-event analysis, was a composite of death from CV causes, non-fatal myocardial infarction, non-fatal stroke, or hospitalization for HF. At a median follow-up of 3.4 years, the rate of the primary outcome was lower in the treatment arm compared with placebo. The benefit was driven by a statistically significant lower incidence of HF hospitalization with finerenone vs placebo (3.2% vs 4.4%; HR 0.71, 95% CI 0.56–0.90), with no differences in CV death alone. New-onset HF was significantly reduced with finerenone vs placebo (1.9% vs 2.8%; HR, 0.68, 95% CI 0.50–0.93).[113]

A pre-specified individual patient-level, pooled analysis of the 2 trials, including 13 026 patients with diabetic kidney disease, showed a reduction in the composite CV outcome, including CV death, non-fatal stroke, non-fatal myocardial infarction, and HF hospitalizations with finerenone vs placebo (HR 0.86, 95% CI 0.78–0.95). Also, the risk of HF hospitalizations alone was reduced (HR 0.78, 95% CI 0.66–0.92),[119] with consistent benefits across eGFR and/or UACR categories.[113] Thus, the 2023 focused update of ESC HF Guidelines, recommends finerenone for the prevention of HF hospitalization in patients with CKD and diabetes.[68]

In the FINE-HEART study[120] finerenone reduced the risk of hospitalization from heart failure (HR: 0.83; 95% CI: 0.75–0.92; $P < .001$) and the composite kidney outcome (HR: 0.80; 95% CI: 0.72–0.90; $P < .001$).

Thus, patients with type 2 diabetes and CKD (eGFR 25–90 mL/min/1.73 m^2) who have persistent albuminuria (UACR ≥30 mg/g) despite standard of care with glucose-lowering and antihypertensive therapy, including maximally tolerated ACE inhibitor or ARB therapy, and who do not have hyperkalaemia (serum potassium <5 mmol/L), treatment with finerenone is indicated.

Obesity

Obesity is associated with new-onset HF. This relationship is especially evident for incident HFpEF. Indeed, an entity of 'obesity-related HFpEF' has been described.[22] Tackling obesity prior to the onset of HFpEF (i.e. preventing HF in obese individuals) is an attractive strategy. The relationship between obesity and incident HFrEF is less certain. Reducing weight in individuals living with obesity can be achieved by lifestyle interventions (education, diet and exercise), pharmacological approaches or weight loss surgery (*Table 1*).

The LOOKAHEAD trial randomized 5109 patients with overweight or obesity (BMI≥25) and diabetes to either intensive lifestyle intervention or support and education.[23] The intensive lifestyle intervention did not result in a reduction in HF; however, patients with higher baseline fitness and those who achieved sustained improvements in fitness or weight loss exhibited a lower risk of HF, particularly HFpEF, suggesting that improvements in fitness, rather than weight loss alone, are clinically relevant.. Further dedicated trials of lifestyle interventions that achieve greater reductions in body weight and with higher increase in fitness with a primary objective of preventing incident HF events are warranted.

Substantial weight loss in individuals living with obesity can now be achieved by pharmacological strategies. GLP1-RAs and related incretin therapies are being studied in an increasing number of large randomized controlled trials. The largest trial to report on the effect of GLP1-RAs on incident HF was the SELECT trial. Semaglutide reduced the risk of HF hospitalizations with a similar treatment effect in those with and without HF.[21]

Other trials including SURMOUNT-MMO (NCT05556512) in people living with obesity are ongoing. This is not a dedicated HF prevention trial but also includes HF as a component of its primary endpoint as well as a secondary endpoint. Further dedicated trials with a primary objective of reducing incident HF in patients living with obesity are warranted.

It must be underlined that the evidence demonstrating beneficial effects of GLP-1 receptor agonists in HF has primarily been obtained in patients with obesity. The effects of this drug class in lean patients remain unknown and warrant further investigation.

Weight loss can be achieved by a variety of surgical procedures. There are no randomized controlled trials that have investigated whether or not weight loss surgery[121] reduces HF events. In non-randomized, non-blinded, observational studies, lower rates of HF hospitalization have been reported with bariatric surgery. A large ongoing randomized controlled trial of weight loss surgery is being conducted in people living with obesity. The BRAVE trial (NCT05531474) is recruiting 2000 people living with a BMI ≥35. Randomization is to bariatric surgery or usual care (which can include lifestyle interventions and/or pharmacological therapies). HF hospitalization is a component of the primary endpoint as well as a secondary endpoint.

Sedentary habit

Globally, the proportion of people not reaching physical activity targets is increasing. ESC guidelines advise all adults to perform 150–300 min of moderate-intensity physical activity; 75–150 min of vigorous-intensity physical activity, or a combination per week.[15] In 2000, 23.4% of all adults worldwide did not meet these recommendations; in 2020 this has increased to 31.3%. The largest decline in physical activity was seen in women and those >60 years old.[122]

Sedentary lifestyle is an established risk factor for CV events in general,[26,123] but recent data also confirms a higher risk of HF.[124,125] Physical activity mainly declines in young adults as they progress into middle age, and this decrease is associated with higher odds of HF.[124] A dose-response meta-analysis of prospective studies demonstrated a reduction of 11% in HF incidence per 20 metabolic equivalent of task hours (MET-hours) per day of total activity.[125] Similarly, high cardiorespiratory fitness is strongly associated with a lower incidence of HF and again a dose-response effect was observed[126] (*Table 2*).

Of note, the largest reduction of CV events (including HF events) can be achieved by reaching physical activity levels to 8.75 MET hours per week.[25] Achieving the recommended physical activity in all, would prevent 15.7% (95% CI 13.1–18.2) of all premature deaths. When translating this to the popular 'daily step count', incident CV disease can already be prevented by increasing daily step count from 2000 to 2735 steps per day, with an optimal dose at 7126 steps per day.[26]

Hypercholesterolaemia

Statins are highly effective for primary prevention of atherosclerotic cardiovascular disease and mortality. Randomized studies, however, failed to show a protective effect of statins against HF.[27,28] However, a recent large population-based study has shown in individual with atrial fibrillation that statin therapy was associated with a 19% lower risk of HF (hazard ratio, 0.81 [95% CI, 0.78–0.85]).[127] This observation warrants prospective evaluations.

On the other side there is growing interest in the link between blood lipids, particularly lipoprotein(a) (Lp(a)), and HF as a modifiable risk factor. While Lp(a) has been associated with myocardial infarction and aortic stenosis,[128,129] its relationship with HF remains unclear due to conflicting evidence (*Table 2*). Ethnic differences in Lp(a)-related HF risk have also been observed, with stronger associations noted in white participants.[130] Notably, Lp(a) appears to primarily influence HFpEF,[130,131] though some studies report links it to HFrEF.[131] Variability in study methodologies and assays likely contributes to the inconsistencies in findings.

Smoking habit

Smoking is a potent, independent risk factor for HF), exerting its effects through both atherosclerotic and non-atherosclerotic pathways. Recent epidemiological data from 9345 ARIC (Atherosclerosis Risk In Communities) participants without history of HF at baseline indicated that smoking approximately doubles the hazard for developing HF, with similar risk profiles for both HFrEF and HFpEF. There was a dose-response relationship for pack-years of smoking and HF. A more extended period of smoking cessation was associated with a lower risk of HF, but risk remained significantly elevated up to 2 decades.[29]

Alcohol and substance misuse

Substance misuse, including alcohol, cannabis, opioids, and stimulants, is linked to higher HF prevalence and hospitalizations.[132,133] Alcohol intake exceeding 70 g/week increases the risk of HF progression by 4.9 times,[30] while cannabis, opioid, and stimulant users exhibit a 2.52-, 3.48-, and 3.35-fold higher HF prevalence, respectively.[31,32] Moreover, the risk is increasing analysis of HF hospitalization trends over 10 years in the USA showed an increasing prevalence of all forms of psychoactive substances in the HF cohort.[134] Methamphetamine use led to 1.8-fold odds of HFrEF, and methamphetamine-related HF admissions increased from 1.8% to 5.6% from 2009 to 2014.[135]

Substance misuse contributes to ischaemic heart disease (24%)[32] and atrial fibrillation (37%) risk, with mechanisms often unclear, particularly chronic effects. Socio-economic risks and genetic predispositions[136] may further amplify HF risk, highlighting the need for careful evaluation to avoid misattributing HF solely to substance misuse.

Chemotherapy and radiotherapy

Cancer therapy-related cardiovascular toxicity (CTR-CVT) risk may vary according to cancer type and stage, anticancer drugs, doses, and underlying comorbidities. Primary prevention aims to avoid or minimize the development of HF due to therapy in patients without CV disease. CV disease and comorbidities should receive the optimal therapy before and during cancer therapy.

When addressing cancer patients, to prevent CTRCD (Cancer therapy-related cardiac dysfunction) and consequently HF, the first step is to optimize CV risk factor control (e.g. smoking cessation, restriction of alcohol consumption, sedentary habits, obesity), according to the ESC Guidelines.[33]

Another important step is the evaluation of CV risk at baseline, which will categorize patients by low, medium, high and very high risk of CTRCD (*Table 2*). High and very high-risk patients have an indication to minimize the use of cardiotoxic drugs and to start medication with ACE-I/ARB, BB and statins[34] (*Table 2*). HF therapies exhibited conflicting results in CTRCD primary prevention. Atorvastatin 40 mg given for 1 year of treatment, started before anthracycline-based chemotherapy among patients with lymphoma, reduced the chance of a significant decrease in left ventricular ejection fraction compared with placebo.[35] Anticancer effects of SGLT2i shown in preclinical studies along with their well-known cardioprotective effects provided the rationale for investigating their efficacy to prevent cardiotoxicity.[36,37] SGLT2i administration was associated with a significantly decreased risk of developing CTRCD in patients with type 2 diabetes and cancer.[38] In a meta-analysis including 4 observational studies and 5590 patients, all-cause mortality was significantly reduced in patients receiving SGLT2i, as compared to those not on SGLT2i, even though the risk of incident HF and the risk of HF hospitalizations, were not significantly different. In a subsequent study, using the TriNetX Global Research Network, patients with cancer and without previous HF diagnosis, receiving anthracycline therapy were identified and classified into 2 groups based on SGLT2i prescription. Over the 2-year follow-up period, SGLT2i therapy was associated with lower rates of new-onset HF.[39] These preliminary results warrant the design of randomized controlled trials.[40]

From the oncological perspective, some strategies that have been investigated include managing anthracycline-related toxicity by adjusting the infusion time and dose intensity. Dexrazoxane is approved for the prevention of anthracycline-induced cardiotoxicity. The cardioprotective effect is explained predominantly by the ability to reduce oxygen-free radical formation in cardiomyocytes. Dexrazoxane and liposomal anthracyclines are currently approved in patients with high and very high risk.

Infections

Influenza, SARS-CoV-2, and other respiratory infections are associated with an increased risk of HF events (i.e. both incident and worsening HF), through several mechanisms leading to a final common pathway of worsened myocardial contractile function.[137]

The mechanisms of myocardial damage include direct myocardial injury and/or myocarditis, inflammation leading to prothrombotic pathways, destabilization of coronary plaques and acute coronary syndromes, hypoxaemia, fever, sympathetic activation and tachycardia with increased myocardial oxygen demand, inducing ischaemia and triggering arrhythmias.[137] Also, patients with prevalent HF have been shown to experience a much more severe clinical course once infected.[138]

There are no data on the direct beneficial effect of vaccination to prevent HF, but vaccinations may reduce the incidence and/or severity of respiratory infections and prevent HF hospitalization (*Table 2*). In a *post hoc* analysis of the PARADIGM-HF trial, of 8099 study participants with HF, 1769 (21%) received influenza vaccination and this was associated with a reduced risk for all-cause mortality in propensity-adjusted models.[41] The INVESTED trial compared the effect of high-dose trivalent and standard-dose quadrivalent inactivated influenza vaccine in patients with recent HF hospitalization or acute myocardial infarction, with no difference observed in risk of death or cardiopulmonary hospitalization between treatment groups.[42]

Female-specific risk factors

The overall lifetime risk of HF is comparable between women and men, but risk factors, HF phenotypes and prognosis differ significantly.[139] In particular, HFpEF is predominant in women, and it is associated with endothelial dysfunction and chronic systemic inflammation.[68] HFpEF has been attributed to several sex differences in epidemiologic factors as well as intrinsic CV function in women, including the higher prevalence of obesity, the steeper increase in arterial stiffness and hypertension prevalence with aging, the higher prevalence of CV organ damage with metabolic risk factors, poorer diastolic myocardial reserve and higher prevalence of microvascular disease.[68,139,140]

Observational epidemiological studies have also documented the independent association of female-specific reproductive factors with increased risk of HF, including early menarche, early menopause, (i.e. before the age of 45 years),[43] recurrent pregnancy loss, nulliparity,[44] hypertensive pregnancy complications[45] and endometriosis.[46] From menarche to menopause and pregnancy, women undergo important changes in sex hormones. Sex hormones affect cardiovascular health, including lipid profiles, response to insulin, BP, vascular reactivity, inflammation, cardiac remodelling, and HF development. In addition, pregnancy involves alterations in cardiovascular hemodynamics, inflammatory, and metabolic disorders. As such, women's reproductive characteristics are implicated in the pathophysiology of HF, and health care providers should investigate them in women's risk stratification (*Table 2*). As for Sleep disorder breathing, obstructive sleep apnoea predicted incident HF in men but not in women (HR 1.13 per 10-unit increase in AHI): AHI > 30 was associated with higher risk to develop HF than AHI <5.[51] Peripartum cardiomyopathy (PPCM) is defined as HFrEF presenting in the last month of pregnancy or in the months following pregnancy in women another known cause of HF.[141] PPCM has a worldwide prevalence of 1 of 2000 births. Growing evidence documents that PPCM is caused by inflammation, unbalanced oxidative stress and generation of anti-angiogenic 16 kDa prolactin.[142] The following risk factors are identified: ethnicity (higher incidence in Africans), genetic variance in dilated cardiomyopathy genes, pre-eclampsia, multi-gestational pregnancies, maternal age <20 or >40 years, smoking, diabetes, hypertension and family history of cardiomyopathy.[142]

Air pollution

Air quality is increasingly recognized as a significant CV risk factor, with short-term exposure to pollutants like PM_{10} and $PM_{2.5}$ linked to acute HF hospitalizations and mortality.[143] Chronic exposure to $PM_{2.5}$, NO_2, O_3, and O_x is associated with a higher risk of myocardial infarction and HF,[47] even at pollutant levels considered acceptable by current standards[48,49] (*Table 2*) $PM_{2.5}$ is the pollutant most strongly tied to HF risk, with cumulative exposure to multiple pollutants compounding this risk[144] Strategies to mitigate HF risk include reducing atmospheric emissions and using air filters, which have shown potential benefits in improving vascular function.[145] Further studies are needed to confirm these interventions' effectiveness.

Socio-economic status

Socioeconomic status strongly influences HF risk, with factors such as food insecurity, poverty, and social isolation playing significant roles.[50] Food insecurity, particularly prevalent among ethnic minorities in the U.S. (8%–20%), increases HF risk through poor nutrition, including deficiencies in essential vitamins and minerals.[146,147] Adults facing food insecurity have a 1.5-fold greater risk of HF, largely driven by an elevated likelihood of coronary heart disease, a primary cause of HF.[148,149] Limited health literacy,[150] barriers to preventative care and challenges in hypertension detection, treatment adherence and longitudinal follow-up may further amplify this risk in socio-economically disadvantaged populations.[151,152] (*Table 2*).

Social isolation also significantly contributes to HF morbidity, mortality, and healthcare costs. Factors such as female sex, severe symptoms, stress, and fatigue contribute to higher perceived isolation. These factors exacerbate HF risk by increasing inflammation, impairing cardiac function, and delaying necessary medical interventions.[153,154] Neighborhood-level factors including air pollution, restricted access to healthy foods, space for physical activity and transportation may compound these effects by reinforcing adverse cardiometabolic behaviours and limiting engagement with healthcare services.[155,156]

Public health strategies should prioritize addressing social isolation and food insecurity as part of comprehensive HF prevention programmes while also addressing broader socio-economic barriers, identifying at-risk individuals through routine assessment of social determinants of health, leveraging community resources and social prescribing, and using telehealth and virtual support platforms may help to mitigate these upstream drivers of HF, particularly among socially, economically or geographically vulnerable populations.

Subtypes of HF: HFpEF and HFrEF

Although HF prevention as a global entity is essential, specific attention should be devoted to the prevention of the two main HF subtypes: HFpEF and HFrEF. For both the HF subtypes there is often an overlap of risk factors, which explains why no specific predictive scores for HFpEF and HFrEF are currently available.

Arterial hypertension represents a major risk factor for both forms of HF (HR 1.18 CI 95% 1.13–1.24).[2] Diabetes mellitus represents a risk factor for both HF subtypes, with HFpEF being the most frequent phenotype. The relative risk for diabetic patients was 2.06 (CI: 1.73–2.46) with a reported J-shaped association with nadir around 90 mg/dL and increased risk even within the pre-diabetic blood glucose range.[5] Other conditions associated with an increased risk of HFpEF include female-specific risk factors, such as early menopause, recurrent pregnancy loss, nulliparity, hypertensive pregnancy complications, and endometriosis. CKD is a risk factor for both HFpEF and HFrEF. However, in a study by Nisha Bansal *et al.*[17] patients with CKD were more likely to develop HFpEF than HFrEF (HR 2.06, 95% CI 1.66–2.56 and HR 1.80, 95% CI 1.36–2.38, respectively). Obesity and, probably better, adiposity is also a risk factor for HF and is more strongly associated with HFpEF than with HFrEF. In the study by Nazir Savji *et al.*[157] higher BMI was associated with a greater risk of HFpEF compared with HFrEF (HR 1.34 per 1-SD increase in BMI; 95% CI 1.24–1.45 vs HR 1.18; 95% CI 1.10–1.27). Recent analyses show an almost universal prevalence of adiposity, i.e. visceral obesity, in patients with HFpEF and tend to show that adiposity may be the single by far most important cause of HFpEF[158,159,160,161] Although air pollution is associated with an increased risk of coronary artery disease[158] through plaque destabilization and myocardial infarction, this risk factor appears to be more strongly associated with HFpEF than with HFrEF.[162–164] In a recent meta-analysis,[165] five high-risk comorbidities for the development of HFpEF were confirmed: atrial fibrillation (HR 2.92, 95% CI 1.94–4.37), arterial hypertension (HR 2.28, 95% CI 1.35–3.84), diabetes mellitus (HR 1.88, 95% CI 1.54–2.30), obesity (HR 1.70, 95% CI 1.45–2.30), and a history of myocardial infarction (HR 1.62, 95% CI 1.18–2.23). In contrast, for HFrEF, the most important risk factors, beyond those already mentioned above (arterial hypertension, diabetes mellitus, CKD, and obesity), include alcohol and/or substance abuse, exposure to chemotherapy and/or radiotherapy, dyslipidemia, and a history of chronic ischaemic heart disease.

Conclusions

Despite the presence of several gaps in knowledge (*Table 5*), the most effective strategy to reduce the mortality and morbidity associated with HF lies in its prevention. Prioritizing the management of traditional CV risk factors—such as hypertension, CKD, and diabetes—and initiating early treatment where appropriate remains essential. To further alleviate the HF burden, a broader range of risk factors must also be considered. Novel conditions have emerged recently including sex-specific factors, air pollution, socioeconomic status, and infections.

To achieve effective HF prevention, it is essential to identify individuals at high risk in whom preventive strategies should be preferentially implemented. This population includes subjects with multiple cardiovascular risk factors (such as arterial hypertension, diabetes mellitus, obesity, and CKD) and/or those classified as high risk according to currently available risk prediction models (e.g. PCP-HF, PREVENT). Advised follow-up strategies for these high-risk individuals include regular monitoring and optimal management of cardiovascular risk factors, as well as the promotion of lifestyle modifications, in accordance with current ESC guidelines.

It is advised for patients at risk of developing HF, NP[164] biomarker-based screening followed by team-based care, including a CV specialist optimizing guideline-directed medical therapy, can be useful to prevent the development of left ventricular dysfunction (systolic or diastolic) or new-onset HF. It highlights the need for further studies to determine cost-effectiveness and risk of such screening, as well as its impact on quality of life and mortality rate.

Supplementary data

Supplementary data are not available at *European Heart Journal* online.

Declarations

Disclosure of Interest

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Data Availability

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